



THE TECHNICAL UNIVERSITY OF KENYA

Halle Selassie Avenue, P. O. Box 52428, Nairobi, 00200, Tel: +254 (020) 343672, 2219929, 0732 388765
Fax: 2219689, E-mail: examinations@tukenya.ac.ke, Website: www.tukenya.ac.ke

Examinations Office

Our Ref: TU-K/SAR- E & C/PS/004/2022/12 (0040)

January 4, 2023

TO WHOM IT MAY CONCERN

Dear Sir/Madam

SUBJECT: PROOF OF STUDY – KAGWI PHILIP NJOROG

We wish to confirm that **Mr. Kagwi Philip Njoroge** studied Bachelor of Engineering in Mechanical Engineering at the Technical University of Kenya, a public university in Kenya (formerly, The Kenya Polytechnic).

The Technical University of Kenya was established through the elevation of the Kenya Polytechnic University College to full university status in 2012, in line with the provisions of the Universities Act, 2012. The charter was granted in January 2013.

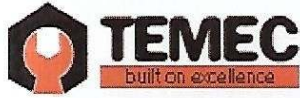
He was admitted to the course in the year 2016 under student registration number **EMMQ/01244/2016** and was awarded a Bachelor of Engineering in Mechanical Engineering – **Second Class Honours (Upper Division)**, in December 2022. All examinations related to the completion of this course were undertaken at the University and the course was taught entirely in **English**.

Should you have any further questions as regards **Mr. Kagwi Philip Njoroge**, kindly contact the undersigned or the office of the Academic Registrar.

Ms. Lucy A. Ogutu

Ag. Senior Assistant Registrar (Examination & Certification)





TEA MACHINERY AND ENGINEERING COMPANY LTD

P.O Box 30213- 00100 Nairobi.

Tel: +254-734-333338/0743031247

Email: info@temec.co.ke

4th May, 2022.

Dear Sir/Madam,

**RE: INDUSTRIAL ATTACHMENT RECOMMENDATION FOR MR. KAGWI
PHILIP NJOROGI (Reg. No. EMMQ/01244/2016).**

This is to acknowledge that Mr. Kagwi Philip Njoroge of Technical University of Kenya (TUK) underwent a 12 weeks Industrial attachment at Tea Machinery and Engineering Company Ltd, (TEMEC) from 1st February, 2022 to 30th April, 2022.

During his attachment Mr. Kagwi was exposed to the following workshop processes.

- Lathe Machine Operation
- Universal Milling Machine Operation
- Shaper Machine Operation
- Shearing machine operation
- Rolling Machine Operation
- Flange rolling Machine operation
- Different welding techniques
- Belt vulcanizing processes
- Dynamic balancing machine operation
- Electrical/Mechanical plant maintenance
- Basic AutoCAD/ Inventor 2D/3D Design & Drafting

Mr. Kagwi exhibited a high level of discipline and cooperation in all stations he was attached to and was able to undertake basic tasks in the workshop with minimum supervision.

The three months industrial experience Mr. Kagwi was exposed to at Temec equips him with basic knowledge in a wide range of Tea Processing Machinery, production techniques and basic AutoCAD/ Inventor drafting.

Based on his conduct during attachment at Temec, I highly recommend him to his academic supervisors and prospective employers as a hardworking, honest and diligent young man able to work with minimum supervision.

Yours Sincerely

Robert Kasamu.
Workshop Manager

Monday 23rd May 2016

PHILIP NJOROGI KAGWE RECOMMENDATION

Philip Njoroge joined Alliance High School in February 2012, after emerging the top student in the country in the National primary exams (8th Grade) KCPE of 2011. I have known Njoroge for four years as a student in my Electricity class and P.E. (Physical education). He struck me as a quiet and obedient young man, but turned to be a disciplined, determined and brilliant student whose quest for knowledge and research abilities lead him to join and become an active member of the science club and Robotics club, and electricity club. I have been a teacher of Electricity and Physical Education for 20 years and I taught Njoroge Electricity and Physical Education for his entire four year stay at the school.

Njoroge was an active member of the aviation club, science club and the coordinator of the electricity club. By researching on a wide variety of topics he was able to present a topic on bistable controls in electricity and in agriculture. That led him to participate in interclass and represent the school in interschool Engineering Fair. In that contest, his performance was outstanding attaining position 3 at the Kikuyu Sub – County level. His quest for knowledge led him to make frequent visits to the electricity workshop to design and make projects that had diverse applications. At one time Njoroge made an amplifier while in form three. During a research visit by Drexel University students, he was selected to participate in a bio – filtration project involving water plates such as water lilies and algae to purify water.

Njoroge was active in math club, where he participated in interhouse math contest and emerged position 3 in year 2012, position 2 in 2013, and position 3 year 2014. He also took part in interclass form 1 drama and his class was impressive, inter – house drama, music and elocution. He participated in Inter – house cross country, athletics and standards, and lawn tennis. Despite this busy schedule in sports and academic contests, he was able to strike a balance and maintain very good grades in his academics.

Njoroge was an active chess player and participated in various chess tournaments and performed very well. He represented the school in the World Youth Chess regional qualifiers at Makini school and emerged position 15.

In his quest to give back to the society, he joined the Red Cross Club where he did a first aid training course and participated in cleaning up Kikuyu town, he was a peer-counselor in class, and attended Christian union rally. He was an assistant table leader in the dining hall. As part of preservation of the environment, he planted trees and flowers every planting season at school and participated in farm work program which involves caring for tree seedling and plucking the under grown shrubs on the house paddock. Despite this busy schedule, he was able to strike a balance between his co-curricular activities and academic work and emerge top of his class with a mean grade 'A' at KCSE exam.

This is why I strongly believe he will be an asset in your institution who will enrich and add value to the campus. His only weakness is his determination to succeed and presentation skills.

MAHINDA NDERITU

Electricity Teacher and Director of Sports



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P. O. 52428,

Nairobi, 00200

RESULT SLIP

This Result slip is a provisional academic statement for the ended semester/term.

STUDENT NAME KAGWI PHILIP NJORGE **REG. NO.** EMMQ/01244/2016
FACULTY OF ENGINEERING AND THE BUILT ENVIRONMENT
PROGRAMME NAME BACHELOR OF ENGINEERING MECHANICAL **PROGRAMME CODE** EMMQ/2016
ENGINEERING
SEMESTER 1 & 2 **ACADEMIC YEAR** YEAR 5

COURSE CODE	COURSE NAME	GRADE
EEMQ5131	MECHATRONIC SYSTEMS DESIGN	B
EEMQ5262	ALTERNATIVE ENERGY ENGINEERING	A
EEMQ5203	LAW FOR ENGINEERS	B
EEMQ5164	ENVIRONMENTAL POLLUTION AND CONTROL	B
EEMQ5291	ENGINEERING PROJECT B	C
EEMQ5163	BUILDING SERVICES ENGINEERING	A
EEMQ5132	INDUSTRIAL PROGRAMMABLE LOGIC CONTROLLERS	A
EEMQ5261	POWER PLANT ENGINEERING	A
EEMQ5202	PROFESSIONAL ENGINEERING PRACTICE	A
EEMQ5191	ENGINEERING PROJECT A	B
EEMQ5162	PLANT ENGINEERING FUNDAMENTALS	B
EEMQ5391	INDUSTRY BASED LEARNING – EXTERNAL ATTACHMENT	A
EEMQ5263	REFRIGERATION AND AIR-CONDITIONING	B
EEMQ5161	INDUSTRIAL ENERGY SYSTEMS AND MANAGEMENT	B
EEMQ5201	TECHNOLOGY AND ENGINEERING MANAGEMENT	A

REMARKS

PASS

Director of School

for: Signature

28/10/2022
Date



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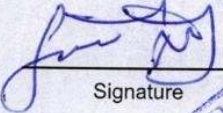
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STUDENT NAME KAGWI PHILIP NJOROGI **REG. NO.** EMMQ/01244/2016
FACULTY OF ENGINEERING SCIENCES AND TECHNOLOGY
PROGRAMME NAME BACHELOR OF ENGINEERING MECHANICAL **PROGRAMME CODE** EMMQ/2016
ENGINEERING
SEMESTER/TERM 1 & 2 **ACADEMIC YEAR** YEAR 4

COURSE CODE	COURSE NAME	GRADE
EEMQ4242	DYNAMICS AND VIBRATIONS OF MACHINES	B
EEMQ4391	INDUSTRY BASED LEARNING – INTERNAL ATTACHMENT II	A
EEMQ4244	GAS DYNAMICS AND JET PROPULSION	B
EEMQ4145	COMPUTER AIDED DESIGN AND MANUFACTURING	B
EEMQ4143	APPLIED HYDRAULICS AND PNEUMATICS	A
EEMQ4241	FINITE ELEMENT ANALYSIS	C
EEMQ4101	ENGINEERING ECONOMICS A	A
EEMQ4141	SOLID AND STRUCTURAL MECHANICS II	B
EEMQ4201	ENGINEERING ECONOMICS B	B
EEMQ4243	JIGS AND TOOLS DESIGN	A
EEMQ4147	INTERNAL COMBUSTION ENGINES	B
EEMQ4232	SAFETY AND RISK ENGINEERING	A
EEMQ4146	MAINTENANCE ENGINEERING	A
EEMQ4144	HEAT AND MASS TRANSFER	C
EEMQ4142	GEARS AND GEAR-TRAINS	B
EEMQ4231	MECHATRONICS SENSOR AND ACTUATORS SYSTEMS	B
EEMQ4245	DESIGN OF MACHINE ELEMENTS	B

REMARKS

PASS

Director of School		Date
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ENGINEERING
SEMESTER/TERM 1 & 2 **ACADEMIC YEAR** YEAR 3

COURSE CODE	COURSE NAME	GRADE
EEMQ3242	MECHANICS OF MACHINES IB	A
EEMQ3241	SOLID AND STRUCTURAL MECHANICS IB	C
EEMQ3211	PROBABILITY AND STATISTICS	B
EEMQ3131	CONTROL AND INSTRUMENTATION ENGINEERING	B
EEMQ3111	LINEAR ALGEBRA	C
EEMQ3244	ENGINEERING DESIGN IB	B
EEMQ3231	ROBOTICS AND CYBERNETICS	C
EEMQ3142	MECHANICS OF MACHINES IA	C
EEMQ3212	NUMERICAL METHODS	A
EEMQ3112	PARTIAL DIFFERENTIAL EQUATIONS	B
EEMQ3243	ENGINEERING THERMODYNAMICS IB	B
EEMQ3143	ENGINEERING THERMODYNAMICS IA	B
EEMQ3132	INTRODUCTION TO MECHATRONICS	A
EEMQ3144	ENGINEERING DESIGN IA	B
EEMQ3141	SOLID AND STRUCTURAL MECHANICS IA	C
EEMQ3391	INDUSTRY BASED LEARNING - INTERNAL ATTACHMENT I	A
EEMQ3246	PRINCIPLES OF AUTOMOTIVE TECHNOLOGY	B

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PROGRAMME NAME BACHELOR OF ENGINEERING MECHANICAL ENGINEERING **PROGRAMME CODE** EMMQ/2016
SEMESTER/TERM 1 & 2 **ACADEMIC YEAR** YEAR 2

COURSE CODE	COURSE NAME	GRADE
EEMQ2131	ELECTRICAL ENGINEERING PRINCIPLES	A
EEMQ2144	FLUID MECHANICS A	A
EEMQ2145	WORKSHOP TECHNOLOGY AND PRACTICE A	C
EEMQ2111	ENGINEERING MATHEMATICS II A	B
EEMQ2232	ELECTRICAL MACHINES	B
EEMQ2132	INTRODUCTION TO MATERIAL SCIENCE	B
EEMQ2242	FLUID MECHANICS B	C
EEMQ2211	ENGINEERING MATHEMATICS II B	A
EEMQ2241	ENGINEERING MECHANICS B	C
EEMQ2143	ENGINEERING MECHANICS A	C
EEMQ2231	ELECTRONICS AND MICROPROCESSORS	B
EEMQ2243	WORKSHOP TECHNOLOGY AND PRACTICE B	A
EEMQ2141	COMPUTER AIDED DRAWING AND DESIGN	B
EEMQ2233	ENGINEERING MATERIALS AND METALLURGY	B
EEMQ2221	COMPUTER PROGRAMMING AND SIMULATIONS	A
EEMQ2146	ENGINEERING METROLOGY AND MEASUREMENTS	B

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PROGRAMME NAME BACHELOR OF ENGINEERING MECHANICAL ENGINEERING **PROGRAMME CODE** EMMQ/2016
SEMESTER/TERM 1 **ACADEMIC YEAR** YEAR 1

COURSE CODE	COURSE NAME	GRADE
EEMQ1222	CHEMISTRY B	A
EEMQ1111	ENGINEERING MATHEMATICS IA	A
EEMQ1202	SOCIETY AND CULTURE	B
EEMQ1211	ENGINEERING MATHEMATICS IB	A
EEMQ1123	BIOLOGICAL SCIENCE	A
UCCC1101	COMMUNICATION SKILLS	C
EEMQ1121	PHYSICS A	B
EEMQ1221	PHYSICS B	A
EEMQ1122	CHEMISTRY A	A
EEMQ1124	COMPUTER SCIENCE A	A
EEMQ1141	MECHANICAL ENGINEERING DRAWING A	D
EEMQ1224	COMPUTER SCIENCE B	A
EEMQ1223	EARTH SCIENCE	A
EEMQ1131	INTRODUCTION TO ENGINEERING	C
EEMQ1241	MECHANICAL ENGINEERING DRAWING B	C
UCCC1102	CRITICAL AND CREATIVE THINKING	A
EEMQ1242	INTRODUCTION TO MECHANICAL ENGINEERING	C
UCCC1201	HEALTH EDUCATION	A

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Signature



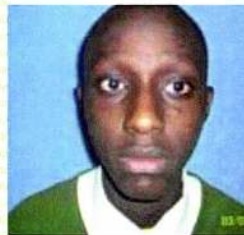
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The Kenya National Examinations Council



KENYA CERTIFICATE OF SECONDARY EDUCATION

This is to certify that the candidate named below sat for the Kenya Certificate of Secondary Education examination in the subjects shown and attained the grades indicated.



11200001/060



NAME: KAGWI PHILIP NJORGE
ALLIANCE HIGH SCHOOL

M1 11200001/060

SUBJECT
101 ENGLISH
102 KISWAHILI
121 MATHEMATICS
231 BIOLOGY
232 PHYSICS
233 CHEMISTRY
313 CHRISTIAN RELIGIOUS EDUCATION
448 ELECTRICITY

GRADE
A- (MINUS)
A
A
A
A- (MINUS)
A
A- (MINUS)
A- (MINUS)



SUBJECTS NAMED EIGHT MEAN GRADE A
EXAMINATION OF YEAR 2015
PRINTED : 161013:12340937

Chief Executive Officer
Kenya National Examinations Council



Chairman
Kenya National Examinations Council

This is a secure document printed using special paper and inks.
Please hold it up to the UV light to verify that the word BARAZA LA MITIHANI and the "ORIGINAL" embedded thread can be seen through the paper.
Not valid without a hologram.

KCSE/ 15 1556927



(See overleaf for conditions of issue)



ALLIANCE HIGH SCHOOL

P.O. BOX 7, KIKUYU Tel: 0202015026
Email: info@alliancehighschool.ac.ke

LEAVING CERTIFICATE

Names:	Kagwi Philip Njoroge		
Son of	Mr. & Mrs. Kagwi		
DOB	27/10/1997	Location:	Ng'Arua
District	Laikipia	Admission No:	10347
Religion	Christian	Date of Issue:	23-May-16

ENTERED THIS SCHOOL ON 31/01/2012 IN FORM One AND LEFT IN 2015
IN FORM Four AFTER COMPLETING FORM Four

PREVIOUS SCHOOL	Year of Entry	Year of Exit	Classes Attended
Riverbank Academy, Nyahururu	2004	2011	Std 1 - 8
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SUBJECTS PERFORMANCE

English	A-	Agriculture	
Kiswahili	A	Art & Design	
Mathematics	A	Business Studies	
Biology	A	Electricity	A-
Physics	A-	Music	
Chemistry	A	French	
History		German	
Geography		Computer Studies	
CRE	A-	-	

KEY
A - Very Good
B - Good
C - Fair
D - Poor

ACADEMIC COMMENTS

He balanced well his academic work and co-curricular activities

GAMES

He played Chess and Tennis

HEALTH

He was in good health while at School

CLUBS, SOCIETIES, RESPONSIBILITIES

He was a member of the Red Cross Society, Electronics, Science and Integrity Clubs

GENERAL COMMENTS

He is a well disciplined, patient and mature man. He has the ability to do well in his future endeavours

GENERAL GRADE **A**

CHARACTER **A**

PRINCIPAL
ALLIANCE HIGH SCHOOL
P. O. BOX 7 - 00502 KIKUYU
TEL: 020-2015026
FAX: 020-2014424

Official Stamp



Inspiring and empowering
future professionals.

Philip Kagwi

Engineering Virtual Experience Program

Certificate of Completion
December 5th, 2022

Over the period of November 2022 to December 2022, Philip Kagwi has completed practical tasks in:

Energy; Power Generation
Healthcare
Aviation Systems

Marshall Schile
Engineering Strategy
Leader for Developing
Technical Talent

Tom Brunskill
CEO, Co-Founder of
Forage

Enrollment Verification Code Z93qht7Z5mPfcX5c | User Verification Code ede4yQpNva9AVXD | Issued by Forage

GLOBAL DEALER LEARNING

This certificate acknowledges that

Philip Njoroge

has successfully completed:

Technicians for Africa Certificate of Completion

Date:

Monday, May 31, 2021


JEFF FITCH, Manager
Global Dealer Learning



CATERPILLAR®



TT3/MVE/15/0011855

GOVERNMENT OF KENYA

NATIONAL INDUSTRIAL TRAINING AUTHORITY

NATIONAL TRADE TEST CERTIFICATE GRADE III



This is to certify that **PHILIP NJOROGE KAGWI**
Whose photograph appears on this certificate has passed a
MOTOR VEHICLE ELECTRICIAN Grade III Trade Test.
Tested at DON-BOSCO BOYS TOWN (KAREN - NAIROBI)
testing centre, DECEMBER 2015
Registration No. **201513376**

NITA
NATIONAL INDUSTRIAL TRAINING AUTHORITY

Trade Testing Officer

Director General - NITA



9.28.2018

*Science & Engineering
Achievement Certificate*

Presented To

Philip Njoroge

*For Participating in the Drexel GK12/Alliance High School Partnership
BioWall Research Competition*



Robert W. Lander
Presented by Eschel Creshaw



CERTIFICATE OF PARTICIPATION

THIS IS TO CERTIFY THAT Philip Njeroge CLASS 4th ADM NO. 10347

PARTICIPATED IN THE

Science and Engineering fair

HELD AT

ALLIANCE HIGH SCHOOL ON 14TH FEBRUARY 2015

SUBJECT: Agriculture Psn 1



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NAIROBI BOTTLERS LIMITED

J. Oloo

HOD SCIENCE
ALLIANCE HIGH SCHOOL

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PRINCIPAL
ALLIANCE HIGH SCHOOL

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